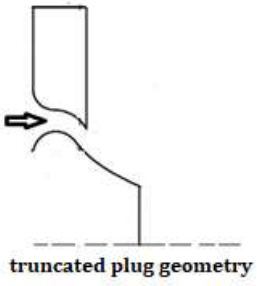
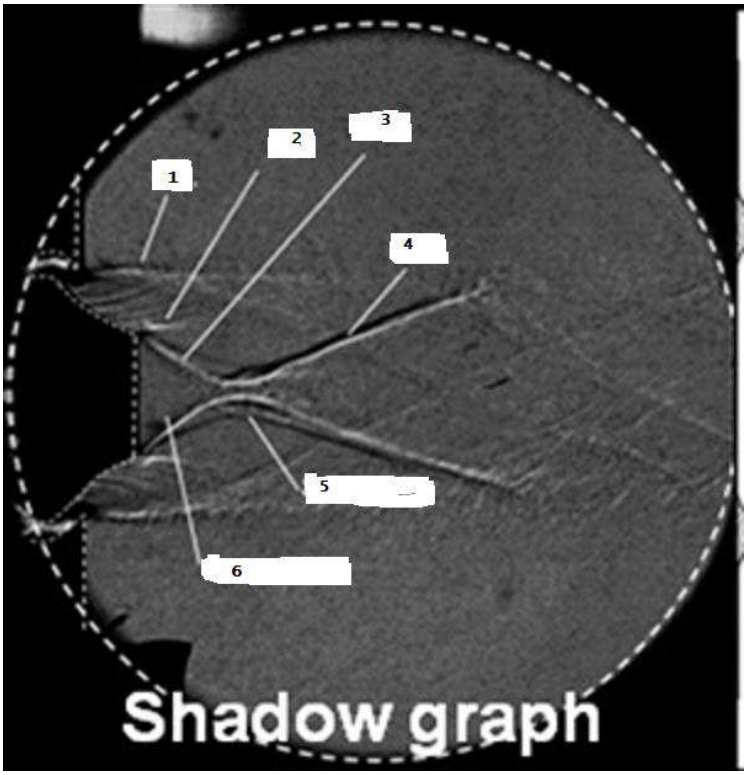
25.01.2023 - Gas Dynamics - MSAR - B. Favini

1. A supersonic steady two-dimensional uniform stream at mach number **M = 1.3** is flowing along a flat plate where impinges an oblique shock. Define the relative pressure-deflection polar and determine the different possible oblique shock reflections (assume ideal gas model). Describe the identified different reflections in the physical (x,y) plane. **(score 13/30)**

2. A piston located at the left boundary of a very long pipe moves to the right with parabolic law $\mathbf{x = t^2}$ up to **t = 1** and after this time with linear law **x = 2t - 1**. The gas inside the pipe is at rest with pressure **p = 1** and density $\boldsymbol{\rho = \gamma}$. Determine the evolution of the field and its dependence on the gas molecular weight (properties $\boldsymbol{\gamma}$ and **R)**. Describe the flowfield on the space-time and pressure-velocity planes. **(score 13/30)**

3. Identify waves and regions numbered in the visualization of the supersonic steady flow generated by a two-dimensional truncated plug nozzle. **(score 4/30)**

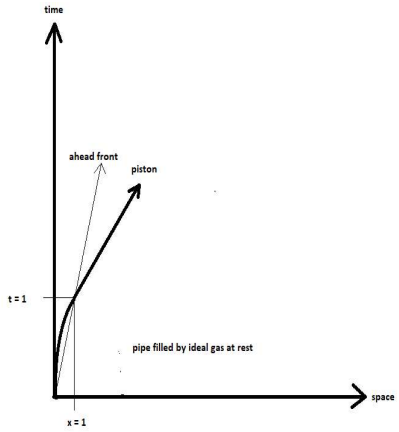


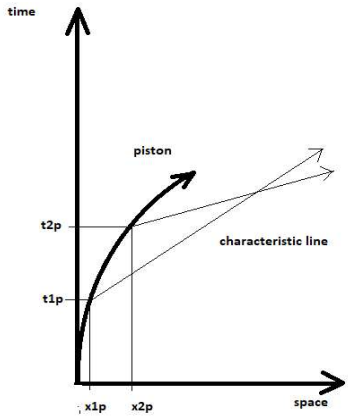


2. Piston propagating in pipe filled by gas at rest

Question:" Determine the evolution of the field and its dependence on the gas molecular weight (properties **γ** and **R)"**

The geometry of the problem on the space-time plane on the left and characteristic coalescence on the right.





The parabolic phase of the piston motion causes the birth of compression simple wave on the u+a characteristic family.

The compression degenerates in a shock by coalescence, i.e. not via a centered compression. The blow-up time is determined as the minimum time for intersecting two adiacent characteristics or, equivalently, when the space derivative of a state variable, velocity for instance, goes to minus infinity (as in the discussion of blow-up via Riemann equation). The first approach, characteristic intersectio, is simpler and requires the construction of the characteristic line equations, characteristics that emanate from the piston.

The characteristic equations are:

$x1 = x1p + (u1p + a1p)\ (t\text{-}t1p)$

$x1 = x2p + (u2p + a2p)\ (t\text{-}t2p)$

where

$x1p = t1p^2$

$x2p = t2p^2$

$u1p = 2t1p$

$u2p = 2t2p$

$a1p = a0 + \delta u1p$

$a2p = a0 + \delta u2p$

with $a0 = 1$

The minimum time for two characteristic lines intersection is the blow-up time and its expression is:

$t_{BU} = 1 / (\gamma+1)$

and the blow-up abscissa is:

$x_{BU} = t_{BU}$

Once the shock occurs its strenght will increase because of the compression wave up to the time when the piston will stop to accelerate, i.e. t=1. Along this time interval, between blow-up time and the start of linear piston law, other than the shock streghtening there occurs the generation on a entropy layer along the particle trajectories and an expansion wave along the u-a characteristic family.

This RCS flow can be determined by the solution of the pseudo-Riemann problem with states:

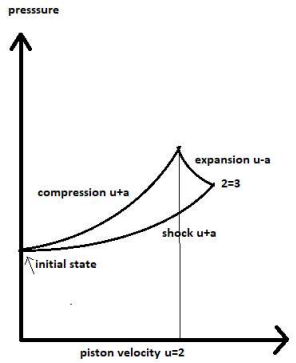
left state: u1=2, p1=γ^(2γ/(γ-1)),a1=γ

(close to the piston at the end of the parabolic phase)

right state: u4=0, p4=1, a4=1

(initial state)

The solution for air gives a shock velocity w=2.87, u2=u3=2.1, p2=p3=9.44, roughly.



This flow it is not the asymptotic state, i.e. the state occurring when the piston moves with constant velocity, because it is necessary that the transient - the interactions between the expansion wave reflected by the piston, the entropy layer and the shock - has ended. This asymptotic flow is equal to the flow generated with an impulsive piston motion except for the existence of the entropy layer, the memory of the transient phase.

Again for air, the asymptotic shock velocity is w=2.76.